

RELATIVE HISTORY

A First-Principles Interpretation of the Parallelogram Law

Why are we reinventing basic geometry?

A conceptual theory of geometry through motion, history, symmetry, and retraceability

1. Why begin again?

Rearranging areas and using dot products are valid methods, but they can feel more consequential than fundamental. They prove that certain geometric relationships are true, while leaving open a different question: why should squared relationships arise so naturally from motion and direction in the first place?

The starting point of this theory is not an area diagram. It is a thought experiment about evidence of motion.

2. A path needs a history

Suppose I texted you and said that I walked straight from point A to point B. If I only sent a picture of myself at A and another at B, that would not prove I walked straight between them. I could have taken a detour. To establish the straight journey, I would need something closer to a video: a record of the intermediate positions between the two endpoints.

A video is made of frames. To translate that into mathematics, suppose the smallest discrete unit of length is epsilon, written ϵ . A path can then be sampled in equal intervals of ϵ .

Simply writing the same interval again and again does not record position:

$$\epsilon + \epsilon + \epsilon + \epsilon + \dots + \epsilon$$

That expression says that equal intervals exist, but not where the traveller was. A positional record instead looks like an accumulation of locations:

$$\epsilon + 2\epsilon + 3\epsilon + 4\epsilon + \dots + L$$

Each term records a later position along the same uniform line.

3. Why the history becomes quadratic

The line is uniform. Equal additional distances produce equal additional changes in position. In functional language, the position map is additive:

$$f(m + n) = f(m) + f(n)$$

If the line contains n equal intervals, then the accumulated positional record is proportional to:

$$1 + 2 + 3 + \dots + n = n(n + 1) / 2$$

For large n , this grows in proportion to n squared. A complete positional record therefore carries a naturally quadratic dependence on the magnitude of the displacement.

Relative history: the amount of positional information needed to record the whole traversal, rather than merely its endpoints.

The file-size analogy is useful here. A photograph of two endpoints contains much less information than a complete video of the path between them. Relative history is the mathematical analogue of that accumulated record.

4. The midpoint as the present

There is another difficulty. Human beings experience a direction of time, but pure numerical data does not inherently know which endpoint is the past and which is the future. The positional data of a journey from A to B can also be read in reverse from B to A.

Instead of privileging one endpoint, begin from the midpoint and expand outward in both directions. The midpoint becomes the present. One orientation is written +A and the reverse orientation is written -A.

The history of a line can therefore be pictured symmetrically around its midpoint:

$$2(\epsilon + 2\epsilon + 3\epsilon + \dots + L/2)$$

This midpoint representation becomes especially useful when several vectors interact. The origin is the natural common present because every midpoint can be translated there without changing the relative structure.

5. Do not choose one future too early

Suppose two vectors A and B meet at the origin. If we immediately replace them by only one resultant, information is lost. A single resultant does not, in general, uniquely reveal both original vectors.

So instead of forcing the system down one path, remain at the common present and keep every possible orientation. For two vectors there are four possibilities:

$$(+A, +B) \quad (+A, -B) \quad (-A, +B) \quad (-A, -B)$$

Geometrically, these directions form an X-like structure around the origin. More generally, if there are n vectors and each has two orientations relative to the present, then there are:

2ⁿ possible past/future combinations

The important object is therefore not one resultant but the complete family of resultants generated by every possible choice of orientation.

6. Overlaying frames

Adding histories is not the same as sequencing them. If the frames of A are followed by the frames of B, the result is merely one path followed by another. A genuinely new resultant direction appears only when information from the two histories is overlaid.

Schematically, groups of frames such as

$$(2\epsilon A + 2\epsilon B) + (4\epsilon A + 4\epsilon B) + (6\epsilon A + 6\epsilon B) + \dots$$

can be compressed into frames belonging to a resultant direction. Several pieces of positional information are then represented inside a single resultant frame, so one resultant viewed alone may appear to contain less information.

The missing information is recovered by the complementary resultant. For example:

$$\frac{1}{2}(A + B) + \frac{1}{2}(A - B) = A$$

$$\frac{1}{2}(A + B) + \frac{1}{2}(B - A) = B$$

The original vectors can therefore be reconstructed when the complementary resultants are kept together. Information should be judged from the complete family of histories, not from one compressed branch.

7. Retraceability and conservation of history

Before any overlay occurs, the system is still at the common present. The histories of A and B remain separately available. No information has yet been compressed into a chosen resultant.

Suppose overlaying two histories changes their apparent relative history by some amount Δ . Now reverse one of the vectors:

$$(+A, +B) \rightarrow (+A, -B)$$

The individual history of A has not changed. The individual history of B has not changed. The same epsilon intervals, the same number of positions, and the same midpoint remain. Only their relative orientation has changed.

The theory therefore treats reversal as reversing the overlay effect itself:

$$\Delta(A, -B) = -\Delta(A, B)$$

$$\Delta(A, B) + \Delta(A, -B) = 0$$

Compression in one particular history is not forbidden. What is forbidden is a net creation or destruction of history after every complementary possibility has been included. Every deviation from the original history appears together with its retraced opposite.

History is conserved because the complete family is retraceable: every interval is still present, and every orientation has its complementary reversal.

8. Why angle is not merely a number

What changes when one vector is reversed? Its magnitude does not change. Its frame count does not change. Its accumulated positional history does not change. The essential change is the angle between the two histories.

If the angle between A and B is θ , reversing B replaces that relation by the complementary angle:

$$\theta \leftrightarrow 180^\circ - \theta$$

This is more than a numerical substitution because an angle represents a position within a rotation. A full turn returns a direction to itself. A half-turn reverses it. A quarter-turn produces a right angle.

$$\text{full turn: } 360^\circ \quad \text{half-turn: } 180^\circ \quad \text{quarter-turn: } 90^\circ$$

The particular number 360 is only a unit convention; the same structure is expressed by 2π , π , and $\pi/2$ radians. What matters is the rotational relationship.

If reversal changes the sign of the overlay effect, then complementary angles satisfy:

$$\Delta(180^\circ - \theta) = -\Delta(\theta)$$

9. Orthogonality as the fixed point of reversal

At 90 degrees, something special happens. Reversing one vector changes θ into 180 degrees minus θ , but when θ itself is 90 degrees, the relative angle is unchanged:

$$180^\circ - 90^\circ = 90^\circ$$

The geometric relation is therefore the same after reversal, even though reversal requires the overlay effect to change sign. Hence the history change at a right angle must satisfy:

$$\Delta(90^\circ) = -\Delta(90^\circ)$$

$$\Delta(90^\circ) = 0$$

Orthogonality is therefore the fixed point of reversal. Angles below 90 degrees and their complements above 90 degrees can carry opposite history effects, but at exactly 90 degrees the relation is self-complementary. No compensation is required: the two histories are already neutral with respect to one another.

10. The parallelogram law

Now consider the two complementary resultants

$$C = A + B \quad \text{and} \quad D = A - B$$

The original histories of A and B are shared between these complementary resultant histories. In the frame-sharing model, each resultant receives half of the relevant frame density. Since relative history itself scales quadratically with magnitude, the comparison of the two sides gives:

$$\frac{1}{2}A^2 + \frac{1}{2}B^2 = \frac{1}{4}C^2 + \frac{1}{4}D^2$$

Multiplying by four gives:

$$2A^2 + 2B^2 = C^2 + D^2$$

This is the parallelogram law. The squared quantities have not been introduced through an area rearrangement or a dot product. They arise from the quadratic accumulation of complete positional history, together with the conservation produced by complementary retraceable histories.

In the orthogonal special case, the complementary resultant magnitudes coincide in the way required to recover the Pythagorean relationship.

11. Many vectors and the sign cube

For n vectors, every vector has two possible orientations relative to the present. The complete family therefore contains 2^n possible histories.

For every configuration containing +B, there is a paired configuration containing -B while all other choices are held fixed. The same elementary intervals remain present; only the relative orientation changes. The history effect of the first configuration is therefore paired with its opposite in the second.

Across the complete family, all such interaction effects cancel in complementary pairs. What remains is the collection of the individual quadratic histories. This is the intuitive structure behind a Rademacher-Parseval type identity: averaging over every sign choice removes cross-interactions while preserving the sum of the separate squared histories.

12. The larger idea

The central claim is not that every resultant preserves history individually. A single resultant may compress several histories into one direction. The claim is that the complete set of possible histories preserves the information of the original vectors.

The present is the state in which nothing has yet been discarded. Choosing one future compresses information. Keeping every possible future restores it. Reversing one history reverses the corresponding interaction. A full rotation restores direction, a half-turn reverses direction, and a right angle is the unique relative angle left unchanged by that reversal.

Because complete positional history grows quadratically with displacement, the quantity preserved by this symmetry naturally takes the form of squared magnitude.

When every interval, every orientation, and every retraceable history is accounted for, total relative history cannot change.

From this perspective, the parallelogram law is not merely an algebraic identity. It becomes a statement about preserving the complete information of motion when all complementary histories are retained.